

PFAS GROMACS Scaffold — Quick Start

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This scaffold runs PFAS and organic-molecule simulations in TIP3P water using OPLS-AA force field parameters generated by LigParGen. Each simulation is isolated in its own directory so multiple systems can coexist without overwriting each other.

Prerequisites

Tool	Purpose
GROMACS (<code>gmx</code>)	All MD steps
Packmol	Recommended solute placement
Python 3	Pre-flight validator
<code>qgmx</code> (<i>optional</i>)	HPC cluster submission

Step 1 — Get LigParGen files

For each molecule, generate a `.gro` coordinate file and an `.itp` topology file from LigParGen. Place them anywhere accessible; you will record the paths in `molecules.csv`.

Step 2 — Initialize a simulation directory

```
scripts/sim_init.sh pfoa-water-001
```

This creates `simulations/pfoa-water-001/` with the required subdirectories and a starter `README.md`. All subsequent commands are scoped to this directory via `SIM_DIR`.

Step 3 — Populate inputs

1. Copy your LigParGen files into `simulations/pfoa-water-001/inputs/ligpargen/`.
2. Edit `simulations/pfoa-water-001/inputs/molecules.csv`:

```
name,resname,gro,itp,count,charge
PFOA,PFO,inputs/ligpargen/pfoa.gro,inputs/ligpargen/pfoa.itp,40,-1
```

Column definitions:

Column	Meaning
name	Molecule name matching [moleculetype] in the .itp
resname	Residue name in the .gro
gro	Path to coordinate file (relative to SIM_DIR)
itp	Path to topology file (relative to SIM_DIR)
count	Number of molecules to place
charge	Expected charge per molecule (used for net-charge check)

Step 4 — Validate inputs

`scripts/validate_gromacs_inputs.py`

Checks that all files exist, molecule names match the .itp, and net charge is sensible before spending time on a build.

Step 5 — Build solute coordinates

Recommended (Packmol):

```
SIM_DIR=simulations/pfoa-water-001 \
BOX_X=8 BOX_Y=8 BOX_Z=8 \
scripts/01_build_with_packmol.sh
```

Alternative (GROMACS-only):

```
SIM_DIR=simulations/pfoa-water-001 \
BOX_X=8 BOX_Y=8 BOX_Z=8 \
scripts/01_alt_build_with_insert_molecules.sh
```

Output: `simulations/pfoa-water-001/build/system_unsolvated.gro`

Step 6 — Solvate and add ions

```
SIM_DIR=simulations/pfoa-water-001 \
ION_CONC=0.15 \
scripts/02_solvate_and_ionize.sh
```

Adds TIP3P water and neutralizing NaCl at the specified concentration. Output: `simulations/pfoa-water-001/bu`

Step 7 — Run the MD pipeline

Local (default):

```
SIM_DIR=simulations/pfoa-water-001 \  
scripts/03_run_md_pipeline.sh
```

HPC cluster via **qgm**x:

```
SIM_DIR=simulations/pfoa-water-001 \  
USE_QGMX=1 NPROCS=8 \  
scripts/03_run_md_pipeline.sh
```

Runs four stages in sequence:

Stage	MDP file	Description
em	minim.mdp	Energy minimization (steepest descent)
nvt	nvt.mdp	NVT equilibration, 100 ps, V-rescale thermostat
npt	npt.mdp	NPT equilibration, 100 ps, C-rescale barostat
production	md.mdp	Production run, 500 ps

Production trajectory: `simulations/pfoa-water-001/runs/production/production.xtc`

Environment variables

All tunable parameters are documented in `.env.template`. Copy it to `.env` and `source .env` before running, or set variables inline as shown above.

Key variables:

Variable	Default	Purpose
SIM_DIR	project root	Active simulation directory
GMX	gmx	GROMACS executable
BOX_X/Y/Z	6 nm	Box dimensions
ION_CONC	0.15 M	Salt concentration
USE_QGMX	0	1 to use qgm x cluster wrapper
NPROCS	1	Processor count for qgm x
MAXWARN	0	Allowed grompp warnings

Running a second simulation

Each simulation directory is fully independent:

```
scripts/sim_init.sh pfos-water-001  
# populate simulations/pfos-water-001/inputs/  
SIM_DIR=simulations/pfos-water-001 scripts/01_build_with_packmol.sh
```

```
SIM_DIR=simulations/pfos-water-001 scripts/02_solvate_and_ionize.sh  
SIM_DIR=simulations/pfos-water-001 scripts/03_run_md_pipeline.sh
```

Neither run touches the other's files.

Provenance

Every script run appends a timestamped record to `$SIM_DIR/sim.log` capturing the script name, key variables, GROMACS version, and Python version. Commit `sim.log` with your simulation directory so the record travels with the data.

Further reading

- `docs/gromacs-ligpargen-template-plan.md` — full workflow reference and failure modes
- `docs/multiple-simulations-management-plan.md` — naming conventions and directory layout
- `docs/packmol-solute-workflow.md` — detail on the Packmol solute-placement path